

cult.fit: System Architecture Document For Integrated Health & Wellness Platform

Anjan Mahapatra

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Abstract

This document presents the comprehensive system architecture for **cult.fit**, India's leading full-stack health and wellness platform offering physical fitness (cult gyms), digital workouts (cult.live), nutrition (eat.fit), mental wellness (mind.fit), and primary healthcare (care.fit). The architecture is designed to support 3.2M+ active customers, 320+ experience stores, 1.5M+ daily active users, 500,000+ monthly meal orders, 30,000+ monthly mental health sessions, and 12,800+ crore annual revenue by FY30F.

The architecture follows a microservices-based, cloud-native design with real-time workout streaming, IoT-enabled gym equipment integration, AI-driven personalisation, and blockchain-verified wellness credentials. Key components include: (1) multi-channel customer onboarding (web, mobile, in-store kiosk), (2) Workout Streaming Engine for live/on-demand classes, (3) Subscription Billing System for hybrid memberships, (4) Nutrition Engine (eat.fit) for meal planning and delivery, (5) Mental Wellness Engine (mind.fit) for therapy matching, (6) Healthcare Engine (care.fit) for teleconsultations, (7) AI Personalisation Engine for workout/food recommendations, (8) Logistics Optimisation for meal delivery, (9) Inventory & Demand Forecasting, and (10) Analytics & ESG Reporting Layer.

The architecture emphasises scalability (horizontal to 10M+ daily API requests), fault tolerance (99.95% uptime, 2-hour RTO), security (AES-256, RBAC, PCI-DSS), and compliance with Indian DPDP Act, clinical data standards, and emerging wellness regulations. This document also includes deployment topologies, disaster recovery procedures, capacity planning, testing strategies, and a full glossary of terms.

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1 Architecture Mind Map



Figure 1: High-Level Architecture Mind Map

2 Architecture Overview

cult.fit's system architecture is built on a microservices-based, cloud-native design deployed on AWS with edge processing for real-time workout streaming, meal delivery logistics, and AI personalisation. The architecture is designed to handle the unique challenges of an integrated health platform: real-time video streaming at scale, personalised meal planning, therapy session scheduling, healthcare consultations, and cross-sell synergies across five verticals.

2.1 Design Principles

1. **Microservices-first:** Loosely coupled services (workout streaming, nutrition, mental health, healthcare, billing) can be scaled independently.
2. **Cloud-native:** Leveraging AWS managed services (EKS, RDS, S3, MSK, Lambda) for reduced operational overhead.
3. **Real-time first:** Sub-2 second latency for live workouts, real-time inventory for meal delivery, instant therapist matching.
4. **AI-first:** Personalised recommendations (workouts, meals, therapists) using collaborative filtering and deep learning.
5. **Security-by-design:** Encryption at rest and in transit, RBAC, comprehensive audit logging for health data (HIPAA-aligned, DPDP compliant).
6. **API-first:** RESTful APIs for B2B corporate wellness programmes, ESG reporting, and third-party integrations.
7. **Omnichannel:** Consistent experience across web, mobile, in-store kiosk, and connected wearables.

2.2 System Boundaries and External Dependencies

The architecture integrates with over 20 external systems:

- Payment gateways: Stripe, Razorpay, UPI (PhonePe, Google Pay, Paytm)
- Logistics partners: Zomato, Swiggy, Dunzo, Delhivery, Shadowfax
- Video infrastructure: Mux (encoding), CloudFront (CDN)
- Telehealth: Vidyo, Zoom for Healthcare, Doxy.me
- Wearables APIs: Apple HealthKit, Google Fit, Samsung Health, Fitbit
- Communication: Twilio (SMS/WhatsApp), SendGrid (email)
- Cloud providers: AWS (primary), with fallback to GCP for disaster recovery

2.3 User Personas Supported

The architecture supports seven primary user personas:

Table 1: User Personas and Capabilities

Persona	Type	Key Capabilities
Retail Customers	B2C	Browse workouts, book classes, order meals, book therapy, consult doctors, track wellness passport, manage subscription
Corporate Clients	B2B	Employee wellness programmes, bulk meal plans, team challenges, ESG reporting
Store Operations	Internal	Gym management, class scheduling, inventory (meal ingredients), customer service
Fitness Coaches	Internal	Live class streaming, schedule management, performance analytics, member feedback
Nutritionists	Internal	Meal plan creation, diet tracking, customer messaging
Healthcare Providers	Partner	Teleconsultation, prescription upload, integration with care.fit
Data API Users	External (corporate)	Aggregated wellness data, anonymised health insights, ESG metrics

2.4 High-Level Architecture Layers

Table 2: Architecture Layers

Layer	Components
Client Layer	Web app (React.js), mobile app (React Native), in-store kiosk (React), corporate API (REST+GraphQL), wearables SDK
API Gateway Layer	Kong / AWS API Gateway – routing, rate limiting (100–10k req/min), JWT/API key auth, observability
Microservices Layer	12+ services: Workout Streaming, Subscription Billing, Nutrition Engine, Mental Wellness, Healthcare, AI Personalisation, Logistics, Inventory, Analytics, Notification, Fraud Detection, Blockchain Wellness Passport
Data Layer	PostgreSQL (operational), TimescaleDB (time-series for workout metrics), Redis (caching), Snowflake (warehouse), S3 (data lake), Hyperledger Fabric (wellness credentials)
External Integrations	Payment gateways, logistics, video CDN, telehealth API, wearables
Deployment & Observability	EKS on AWS (ap-south-1, us-east-1), CI/CD (GitHub Actions + ArgoCD), monitoring (Datadog, Prometheus, ELK)

3 Client Layer

3.1 Customer Web Portal

Technology: React.js + TypeScript + Next.js (SSR/SEO) + WebSocket (Socket.io).

Features:

- **Workout catalogue:** AI-recommended classes, live schedule, on-demand library, instructor profiles.
- **Meal plans (eat.fit):** Personalised meal recommendations, calorie tracking, order delivery, subscription management.
- **Mental wellness (mind.fit):** Therapist directory, session booking (video/chat), mood tracker.
- **Healthcare (care.fit):** Doctor consultation booking, prescription upload, health record access.
- **Unified dashboard:** Wellness score, cross-vertical activity (workouts + meals + therapy + health checks), subscription status.

Performance considerations: Next.js static generation for marketing pages, server-side rendering for personalised dashboards. WebSocket connections for real-time notifications.

3.2 Mobile App (iOS/Android)

Technology: React Native + TypeScript + Redux Toolkit + WatermelonDB (offline first).

Features:

- Live workout streaming with chat, instructor feedback, real-time heart rate (wearables).
- Offline mode: download workouts, cached meal plans, therapist profiles.
- Barcode scanning for eat.fit meal pickup at partner restaurants.
- Push notifications for class reminders, meal delivery, appointment alerts.
- Wearables integration (Apple Health, Google Fit, Samsung Health) – sync steps, calories, sleep.

Offline strategy: WatermelonDB stores up to 500 recent workouts and meal plans. Sync engine reconciles changes when online.

3.3 In-Store Kiosk (cult Gym)

Technology: React.js (touch-optimised) + Electron + RFID (member check-in).

Features:

- Member check-in via QR code / RFID tag → real-time attendance tracking.
- Class booking (same day / advance), waitlist management.

- Nutrition product purchase (protein shakes, snacks) linked to member account.
- Feedback collection after each class (rating, comments).

Deployment: Offline-capable; syncs when store Wi-Fi is available. Uses local SQLite.

3.4 Corporate API Access

Technology: RESTful APIs (OpenAPI 3.0) + GraphQL + OAuth2.

Use cases:

- **Corporate wellness:** Bulk employee enrolment, team challenge creation, aggregated wellness reports.
- **ESG data aggregators:** Scope 3 emissions from employee commute (via gym attendance), health outcomes.
- **Research institutions:** Anonymised fitness/nutrition/mental health trends (academic license).

Rate limits per API key: 5,000 requests per minute for platinum tier, 1,000 for gold, 200 for free academic.

4 API Gateway Layer

Table 3: API Gateway Configuration

Component	Specification
Technology	Kong / AWS API Gateway
Authentication	JWT (customers), API keys (corporate), OAuth2 (B2B), mTLS (internal)
Rate Limiting	200/min (customers), 2k/min (corporate), 50k/min (internal)
SSL/TLS	TLS 1.3 only, auto renewal (ACM)
DDoS Protection	AWS Shield Advanced + WAF (OWASP Top 10)
Request Validation	JSON schema (draft-07), limits: 2MB (customer), 10MB (corporate)
Logging	ELK stack – 90d hot, 1y cold
Caching	Redis cluster (3 shards, 2 replicas) – product catalog, sessions

4.1 API Versioning Strategy

- URI versioning: `/v1/workouts`, `/v2/meals`. Breaking changes → major version.
- Deprecation policy: 6 months notice, 12 months backward compatibility.
- Current versions: v1 (stable), v2 (beta – GraphQL, WebSocket streaming).

4.2 Request Flow and Transformation

The gateway enriches requests with correlation IDs, validates JWTs, applies rate limit counters (stored in Redis), and routes to appropriate microservices. Failed requests are retried up to 3 times with exponential backoff.

5 Microservices Layer

All services communicate via **gRPC** (synchronous) and **Apache Kafka** (asynchronous events). Below is a summary table followed by deep dives into key services.

5.1 Microservices Summary

Table 4: Microservices Overview

Service	Stack	Priority	Function
Workout Streaming	Node.js + Mux + Redis	H	Live/on-demand video, real-time chat, wearables data ingestion
Subscription Billing	Node.js + Stripe + PostgreSQL	H	Hybrid membership (gym+digital), recurring billing, dunning
Nutrition Engine (eat.fit)	Python + FastAPI + PostgreSQL	H	Meal planning, order management, calorie tracking, kitchen integration
Mental Wellness (mind.fit)	Python + PostgreSQL + Twilio	H	Therapist matching, video sessions, mood tracking, crisis alerts
Healthcare (care.fit)	Python + PostgreSQL + HIPAA	H	Teleconsultation, e-prescription, health records, lab integration
AI Personalisation	Python + TensorFlow + Feast	H	Workout/meal/therapist recommendations, churn prediction
Logistics	Go + OR-Tools + PostGIS	M	Meal delivery routing, real-time tracking, courier allocation
Inventory & Forecasting	Python + Prophet + XGBoost	M	Ingredient demand forecast, auto-reorder, waste reduction
Analytics & Reporting	Python + dbt + Superset	M	Executive dashboards, ESG reports, cross-sell analytics
Notification	Node.js + Bull + Resend	L	Email, SMS, push, WhatsApp alerts
Fraud Detection	Python + XGBoost + Seldon	L	Payment fraud, coupon abuse, fake class bookings
Wellness Passport	Go + Hyperledger Fabric	H	Blockchain-verified wellness credentials (workout hours, meal streaks, therapy sessions)

5.2 Deep Dive: Workout Streaming Service

Handles live and on-demand workout video streaming, real-time chat, and wearables data.

- **Technology:** Node.js 20.x, Mux (video API), Redis (presence), WebSocket.
- **API Endpoints (examples):**
 - POST `/v1/stream/start` – instructor starts a live class.
 - GET `/v1/stream/watch/classId` – customer joins stream (HLS URL).
 - POST `/v1/wearables/hearttrate` – ingest heart rate data (validated via JWT).
- **Data flow:** Studio camera → RTMP ingest → Mux transcoding (multi-bitrate) → CloudFront CDN → client HLS player. Wearables data → WebSocket → Kafka topic `wearables.metrics` → consumed by Analytics and AI services.
- **Scaling rules:** HPA based on CPU & WebSocket connections. Target 50,000 concurrent viewers at FY30F.
- **Fault tolerance:** Stateless; auto-scaling group. On failure, new pod joins within 30s.
- **Real-time chat:** Uses Socket.io rooms; messages persisted to PostgreSQL via Kafka for replays.

5.3 Deep Dive: Nutrition Engine (eat.fit)

Manages meal plans, kitchen operations, and delivery logistics.

- **Technology:** Python FastAPI, PostgreSQL, Redis, Celery.
- **Key data entities:** meals, ingredients, orders, delivery_slots.
- **Order flow:** Customer places order → API writes to PostgreSQL → Kafka event `order.created` → Kitchen display system (KDS) → Celery task for courier assignment → Logistics service.
- **Inventory sync:** Real-time ingredient stock across 12 cloud kitchens. When stock $\leq$ threshold, auto-reorder via Inventory Forecasting service.
- **Target volume:** 500,000 daily orders by FY30F.
- **Slot management:** Dynamic pricing for delivery slots; algorithm balances demand and delivery capacity.

5.4 Deep Dive: AI Personalisation Service

Provides cross-vertical recommendations.

- **Technology:** Python, TensorFlow 2.x, Feast (feature store), Redis.
- **Models:**

- Two-tower neural network for workout recommendations (user features + class features).
- Collaborative filtering for meal suggestions (implicit feedback – order history).
- Gradient boosting for therapist matching (based on past session outcomes, mood scores).
- **Training pipeline:** Daily batch job (Spark on EMR) → model registry (MLflow) → A/B testing (10% canary).
- **Inference latency:** <100ms p95, cached in Redis for 1 hour.
- **Feature store:** Feast manages online (Redis) and offline (S3) feature sets.

5.5 Event-Driven Architecture with Kafka

All state changes are published to Kafka topics. Key topics and their consumers:

Table 5: Kafka Topics and Consumers

Topic	Produced By	Consumed By
<code>workout.completed</code>	Workout Streaming	Analytics, Wellness Passport, Notification
<code>meal.ordered</code>	Nutrition Engine	Inventory, Logistics, Analytics
<code>therapy.session-completed</code>	Mental Wellness	Billing, Analytics, AI Personalisation
<code>payment.succeeded</code>	Billing	Subscription, Analytics, Notification
<code>user.registered</code>	Auth Service	AI Personalisation, Analytics, Marketing
<code>fraud.alert</code>	Fraud Detection	Admin, Notification

Consumers can replay topics for debugging or backfill. Kafka retention is 7 days for most topics, 30 days for audit topics.

6 Data Layer

Polyglot persistence for different workloads.

Table 6: Operational Databases

Database	Technology	Purpose (FY30F size)
Primary OLTP	PostgreSQL 15	Customers, orders, subscriptions, payments, health records (25 TB)
Time-series	TimescaleDB	Workout metrics (heart rate, calories), meal delivery tracking (15 TB)
Blockchain	Hyperledger Fabric	Wellness Passport audit trail (10 TB)
Caching	Redis Enterprise	Sessions, catalogue, real-time inventory (500 GB)
Search	Elasticsearch 8.x	Workout search, meal search, therapist search (5 TB)

6.1 Data Warehouse

Technology: Snowflake (auto-scale). Data from Kafka (40+ topics) → dbt (100+ models) → Snowflake.

- Volume: 20 TB raw/month (FY26E) → 100 TB by FY30F.
- Refresh: Real-time (Snowpipe) for operational dashboards; daily batch for analytics.
- Data mesh architecture: Domain-oriented data products (fitness, nutrition, mental health, healthcare) owned by respective teams.

6.2 Data Lake

Technology: Amazon S3 (intelligent-tiering) + Glacier Deep Archive.

- Workout videos (original recordings) – retained 90 days.
- User health data (encrypted, AES-256) – retained 7 years post-last activity (DPDP requirement).
- System logs – 30d hot, 90d warm, 1y cold.
- ML model artifacts – versioned indefinitely.
- Data catalog: AWS Glue with partition evolution.

6.3 Data Replication and Backup

- PostgreSQL: Point-in-time recovery (PITR) with 35-day retention. Cross-region replica to ap-southeast-1.
- TimescaleDB: Continuous aggregates for downsampling older metrics.
- Redis: AOF persistence + daily snapshots to S3.
- Elasticsearch: Snapshot lifecycle management to S3.
- Hyperledger Fabric: Ledger backed up to S3 every hour.

7 Scalability & Performance

7.1 Scaling Targets

Table 7: Horizontal Scaling Targets by FY30F

Metric	FY26E	FY30F
Active customers (monthly)	850,000	3,200,000
Daily active users	250,000	1,500,000
Daily API requests	2,000,000	12,000,000
Daily meal orders	150,000	500,000
Concurrent workout viewers	10,000	50,000
Monthly therapy sessions	30,000	150,000
Data warehouse size (TB)	20	100

7.2 Performance SLAs

Table 8: Performance Service Level Agreements (p95)

Service	Target	Current
Workout streaming (join latency)	<2 sec	1.2 sec
Meal order placement	<1 sec	0.6 sec
Therapist matching	<500 ms	300 ms
AI recommendation	<100 ms	70 ms
Checkout (incl. payment)	<2 sec	1.1 sec
Wellness Passport verification	<500 ms	250 ms

7.3 Capacity Planning Methodology

Capacity reviews are conducted quarterly using historical growth rates, seasonality (Diwali, New Year, summer), and marketing campaign calendars. Autoscaling thresholds are set at 70% CPU or 80% memory for stateless services. Stateful services (PostgreSQL) use read replicas and vertical scaling up to r6i.8xlarge.

7.4 Disaster Recovery & Multi-Region

- Primary region: ap-south-1 (Mumbai). Secondary: ap-southeast-1 (Singapore).
- Active-passive with asynchronous replication (RPO 15 min). Failover via Route53 health checks.
- RTO: 2 hours (1 hour for critical services: billing, streaming).
- Regular chaos engineering (AWS Fault Injection Simulator) – monthly failover drills.
- Data residency: Indian user data remains in ap-south-1; international users can be served from Singapore.

7.5 Load Testing Strategy

Monthly load tests simulate peak traffic (3x normal) using Locust distributed from 10 AWS regions. Tests cover API gateway, all microservices, and database under load. Results are compared against SLOs.

8 Security Architecture

8.1 Authentication & Authorization

Table 9: Authentication Methods by User Type

User Type	Authentication Method
Retail Customers	JWT + password (bcrypt, 10 rounds). 2FA optional. 8 hr timeout.
Corporate Clients	API keys + OAuth2 + IP whitelist. Key rotation 90 days.
Store Associates	JWT + MFA (TOTP). Device attestation.
Fitness Coaches	JWT + biometric (face/touch). Session timeout 4 hr.
Healthcare Providers	OAuth2 + client certs + audit logging.
Admins	MFA (TOTP) + YubiKey. Break-glass requires 2-person approval.
Internal Services	mTLS (Istio service mesh), SPIFFE identities, auto cert rotation.

8.2 RBAC Roles

Table 10: RBAC Roles (8 roles)

Role	Permissions
Customer	Own profile, workout history, meal orders, therapy sessions, health records, subscription. No access to others.
Corporate Client	View employee wellness aggregates, download ESG reports, manage team challenges.
Store Associate	Check-in members, book classes, sell nutrition products, view store inventory.
Fitness Coach	Schedule classes, view enrolled members, stream live workouts, access class analytics.
Nutritionist	Create meal plans, view assigned customer diets, send messages.
Healthcare Provider	View patient assigned appointments, upload prescriptions, update health records.
Admin	User management, system config, feature flags, report access, audit log viewer.
Super Admin	Full access (break-glass only, 2-person approval), DR failover authorisation.

8.3 Data Encryption

- **At rest:** AWS EBS volumes encrypted (AWS KMS). RDS encryption enabled. S3 server-side encryption (AES-256).
- **In transit:** TLS 1.3 for all external APIs. mTLS for inter-service (Istio).
- **Application level:** Sensitive fields (Aadhaar, health records) encrypted using AWS KMS + envelope encryption before storing in PostgreSQL.

8.4 Compliance

- DPDP Act (India) – consent management, data localisation, deletion requests.
- HIPAA-aligned for care.fit (business associate agreements with tele-health providers).
- PCI-DSS Level 1 for payment processing (Stripe handles primary compliance; cult.fit maintains SAQ A).
- ISO 27001 certified data centers (AWS).

8.5 Security Monitoring and Incident Response

- SIEM: Splunk Cloud (aggregates logs from AWS, Kubernetes, applications).
- Intrusion detection: GuardDuty + Falco (runtime security).
- Vulnerability scanning: Amazon Inspector (weekly) + Snyk (container image scanning in CI).
- Incident response: On-call security engineer 24/7. Runbooks for data breach, DDoS, ransomware.

9 CI/CD & Observability

9.1 CI/CD Pipeline

- Source: GitHub (trunk-based development).
- CI: GitHub Actions – unit tests, linting, security scan (Snyk), build container image.
- CD: ArgoCD (GitOps) – syncs from ECR to EKS. Automated canary deployments (10% → 50% → 100%).
- Rollback: Instant via ArgoCD rollback or revert Git commit.
- Approval gates: Production deployments require QA signoff and business hours approval.

9.2 Observability Stack

- **Metrics:** Prometheus + Grafana (node, pod, service metrics). Datadog for business KPIs (orders per second, streaming health).
- **Logs:** ELK stack (Filebeat → Logstash → Elasticsearch). Centralised dashboards with log-based alerts.
- **Traces:** Jaeger (distributed tracing) – sample 5% of requests; trace retention 14 days.
- **Alerting:** PagerDuty for critical alarms (error rate >1%, payment failures). Slack bots for non-critical.
- **SLO tracking:** Error budget dashboards for each microservice.

9.3 Incident Management

- Severity levels: SEV0 (total outage) – 15min response, 1hr mitigation target.
- Post-mortem blameless culture, with action items tracked in Jira.
- ChatOps: ‘/incident’ command in Slack creates a dedicated channel and bridge to PagerDuty.

9.4 Chaos Engineering

Monthly GameDays using AWS Fault Injection Simulator (FIS) to test:

- Pod failures (kill 30% of replicas)
- AZ outage (deprioritise one availability zone)
- Network latency injection (500ms between services)
- Database connection pool exhaustion

Success criteria: no SEV0/SEV1 incidents, auto-recovery within SLOs.

10 Testing Strategy

- **Unit tests:** Jest (Node.js), pytest (Python), go test (Go). Coverage $\geq 80\%$.
- **Integration tests:** Testcontainers for databases, Kafka, Redis. Run in CI.
- **Contract tests:** Pact between producers and consumers of APIs.
- **End-to-end tests:** Cypress for web, Detox for mobile. Run on staging daily.
- **Performance tests:** Locust clusters; run before every major release.
- **Security tests:** OWASP ZAP DAST scan on staging; Snyk container scanning.

11 Deployment Topology

- Kubernetes clusters: 3 production clusters (Mumbai primary, Singapore DR, Bangalore staging).
- Node pools: High-CPU for streaming (c5.4xlarge), high-memory for databases (r5.2xlarge), general for others (m5.2xlarge).
- Service mesh: Istio (traffic splitting, mTLS, circuit breaking).
- Ingress: AWS Load Balancer Controller + Kong.
- Storage: EBS gp3 for stateful workloads, EFS for shared configs.

12 Conclusion

The cult.fit system architecture is designed to support the company's growth from 2,800 crore (FY26E) to 12,800 crore (FY30F) while maintaining high availability (99.95% up-time), sub-second response times for critical operations, and full compliance with Indian DPDP Act, clinical data privacy, and emerging wellness regulations.

12.1 Key Architectural Decisions

Table 11: Key Architectural Decisions & Rationale

Decision	Rationale
Microservices (vs. monolithic)	Independent scaling (streaming vs. billing), team autonomy, fault isolation.
EKS (K8s) over ECS	Portability (multi-cloud), HPA for demand-based scaling, CNCF ecosystem.
PostgreSQL + TimescaleDB	Single DB for transactional + time-series (workout metrics), compression (90% savings).
React Native	Single codebase for iOS/Android, offline support, consistent AR/QR features.
Hyperledger Fabric for Wellness Passport	Immutable audit trail for wellness claims, customer trust, smart contracts for rewards.
Multi-region deployment	Low latency for India (ap-south-1), DR protection, data residency.

12.2 Projected Capacity by FY30F

- Active customers (monthly): 3.2M (3.8x growth)
- Daily active users: 1.5M (6x growth)
- Daily API requests: 12M (6x growth)
- Daily meal orders: 500,000 (3.3x growth)
- Concurrent workout viewers: 50,000 (5x growth)
- Data warehouse size: 100 TB (5x growth)
- Annual revenue: 12,800 crore (4.6x growth)

12.3 Architecture Readiness Checklist

- **Scalability:** Horizontal scaling validated via load tests (10x current load) – Yes
- **Resilience:** Chaos testing (pod failures, AZ outages) – ongoing, 95% success.
- **Security:** Penetration test (annual); no critical findings in last report – Yes

- **Compliance:** DPDP readiness audit completed – Yes
- **Observability:** SLO dashboards for all services – Yes

Glossary

Term	Definition
ARPU	Average Revenue Per User
CaaS	Corporate wellness as a Service
CDN	Content Delivery Network
DPDP	Digital Personal Data Protection Act (India)
EKS	Amazon Elastic Kubernetes Service
ESG	Environmental, Social, Governance
gRPC	High-performance RPC framework
HPA	Horizontal Pod Autoscaler (Kubernetes)
HLS	HTTP Live Streaming
JWT	JSON Web Token
Kafka	Distributed event streaming platform
KDS	Kitchen Display System
mTLS	Mutual TLS
OLTP	Online Transaction Processing
P95	95th percentile latency
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SLO	Service Level Objective
SPIFFE	Secure Production Identity Framework for Everyone
TOTP	Time-based One-Time Password

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